

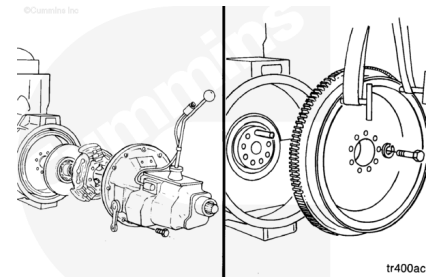
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

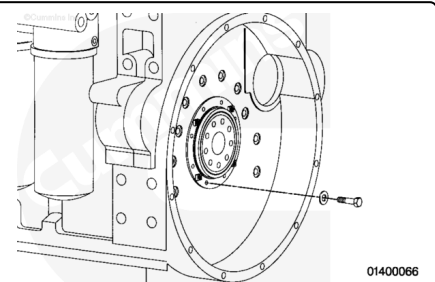
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the transmission, clutch, and all related components. Refer to the equipment manufacturer's instructions
- Remove the flywheel. Refer to Procedure 016-005 in Section (</qs3/pubsys2/xml/en/procedures/20/20-016-005-tr.html>)



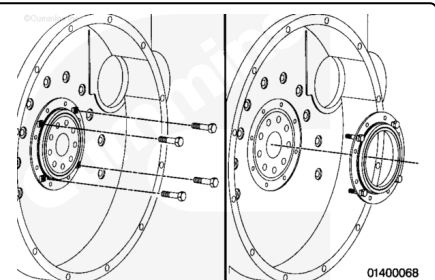
Remove

Remove the seal mounting capscrews.



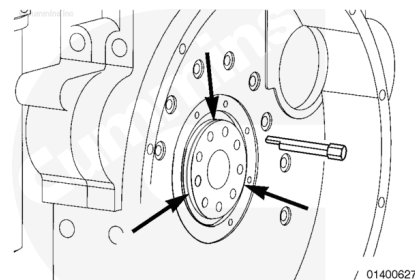
Use four [1/4-20 UNC x 1 inch] capscrews to remove the seal from the crankshaft.

Install the capscrews in the welded nuts in the seal case. Tighten the capscrews alternately and evenly to remove the seal.



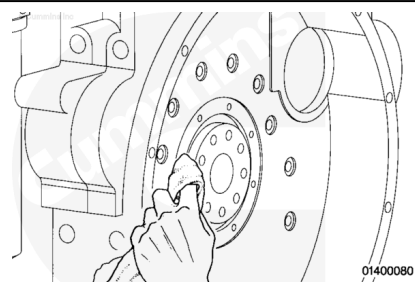
If installed, remove the wear sleeve from the crankshaft using Service Tool Part Number 3824971.

Insert the expander (2) between the wear sleeve and flywheel housing. Turn the expander so the wedge nose deforms the wear sleeve. Repeat this process until the press fit has been reduced and the wear sleeve can be removed from the crankshaft.



Clean

Use a clean cloth and safety solvent, Part Number 3823717, or equivalent, to clean the crankshaft and seal mounting area.

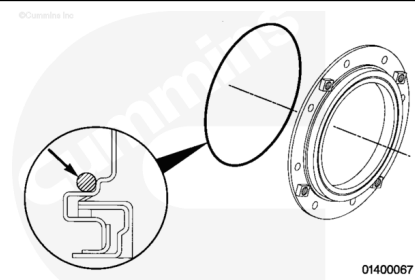


Install

Wet Flywheel Housing

Install a new o-ring on the seal housing.

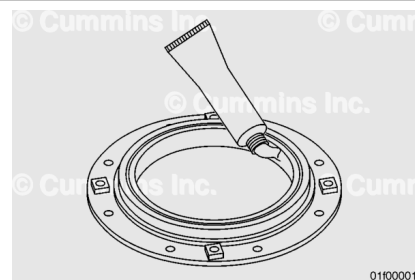
Use vegetable oil to lubricate the o-ring.



Apply a small amount of Lubriplate™ 105, Part Number 3163086, or equivalent, to the inside diameter of the oil seal.

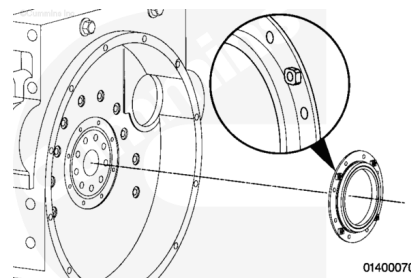
⚠ CAUTION ⚠

Do not allow oil to come in contact with other areas of the seal to avoid damage to the sealing surfaces.

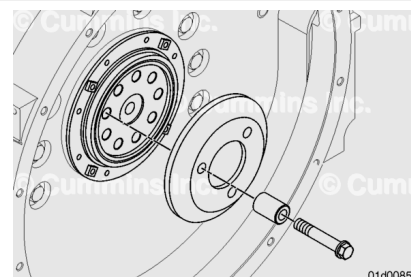


Use the pin supplied in the seal kit to align the seal mounting holes with the mounting capscrew holes in the flywheel housing.

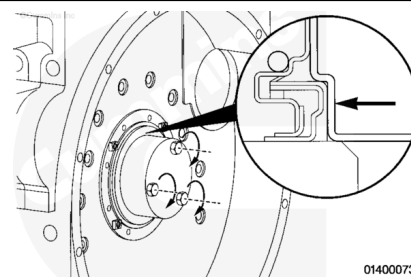
Use hand pressure to push the seal onto the crankshaft as far as possible.



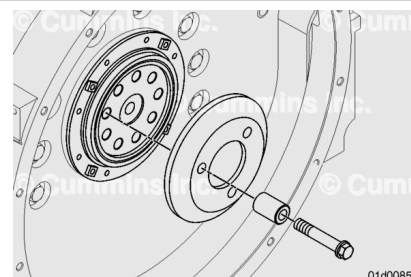
Use three flywheel mounting capscrews to attach the seal installation tool, Part Number 3824971, to the crankshaft.



Tighten the three capscrews alternately and evenly in one-half of a turn increments until the seal carrier seats against the flywheel housing.

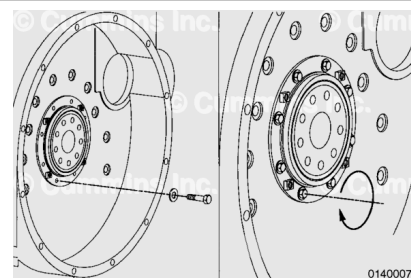


Remove the seal installation tool.



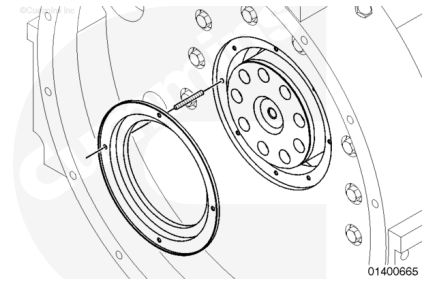
Install the seal mounting capscrews. Tighten the capscrews.

Torque Value: 11 n•m [97 in-lb]



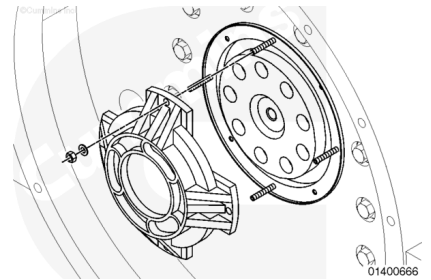
Dry Flywheel Housing

Install the four socket head set screws. Place the seal on the socket head set screws and set the seal squarely against the crankshaft face, pushing **only** against the rubber face.

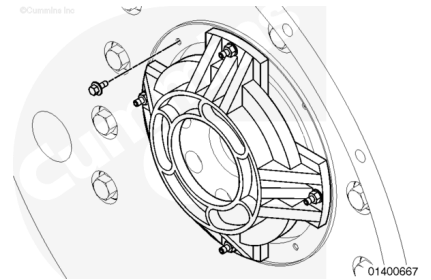


Set the blue colored plastic seal installation tool squarely against the seal face. Make certain that the raised portion of the rear installation tool is facing away from the flywheel housing and that the gray rubber sealing bead is facing toward the flywheel housing.

Place washers and nuts on the four socket head set screws and hand snug evenly.



Tighten the nuts in equal increments (about 2 turns per nut in a criss-cross pattern) until the seal flange contacts the flywheel housing. Install the four new capscrews, Part Number 4922215, and washers (use washers from the old seal capscrews) in the remaining holes and finger-tighten.

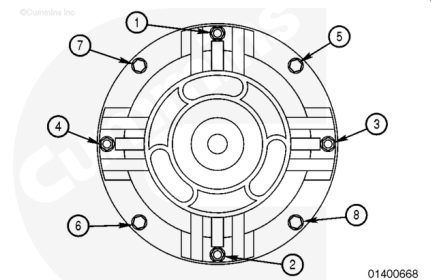


Tighten the four nuts, 1, 2, 3, and 4, against the installer tool in a criss-cross pattern.

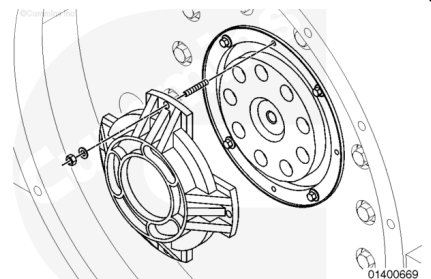
Torque Value: 11 n•m [97 in-lb]

Tighten the four capscrews, 5, 6, 7, and 8, Part Number 4922215, in a criss-cross pattern.

Torque Value: 11 n•m [97 in-lb]

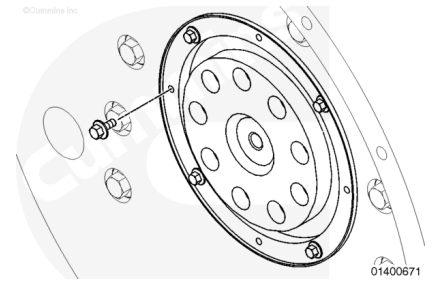


Remove the nuts, washers, plastic installation tool, and the four socket head set screws.



Install the remaining four new capscrews, Part Number 4922215, along with plain washers (use washers from the old seal capscrews). Tighten the capscrews in a crisscross pattern.

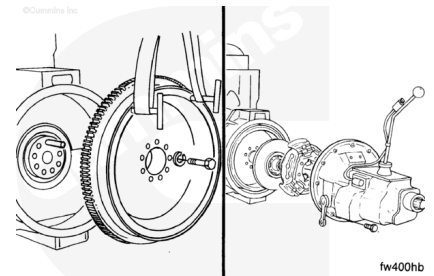
Torque Value: 11 n•m [97 in-lb]



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/20/20-016-005-tr.html>)
- Install the transmission, clutch, and all related components. Refer to the equipment manufacturer's instructions.
- Operate the engine and check for leaks.